Solutions

Q1 Solution

(A)Logistic loss is convex ;; (B)MLE gives cross-entropy ;; (D)decision boundary is linear ;; (E)and zero loss on separable data requires weights to diverge.

Only statement equating logistic loss with squared error is incorrect.

Incorrect option: (C)

Q2 Solution

L_2 regularization keeps the objective convex, prevents weight explosion, reduces variance, and pushes weights toward zero. It does not make the boundary non-linear.

Correct options: (A), (B), (D), (E)

Q3 Solution

At initialization,

$$p = \sigma(0) = 0.5.$$

Gradients:

$$\nabla_w L = (p - y)x = (-0.5)\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0.5 \end{pmatrix}.$$

Sum of weight-gradient components:

$$-1 + 0.5 = -0.5.$$

Bias gradient:

$$\nabla_b L = p - y = -0.5.$$

Hence,

$$S = -0.5 - 0.5 = -1.$$

Final Answer: $S = -1$**.**

Q4

A,C,D,E

Solution to Q5

We compare linear regression and logistic regression for binary classification with labels $y \in \{0, 1\}$.

- **(A) True.** Linear regression minimizes the squared error loss

  $$\ell_{\text{lin}} = (y - \hat{y})^2,$$

  whereas logistic regression minimizes the cross-entropy (negative log-likelihood)

  $$\ell_{\log} = -\left[y \log p + (1 - y)\log(1 - p)\right].$$

- **(B) True.** Logistic regression uses a sigmoid activation, so predicted probabilities always satisfy $p \in [0, 1]$. Linear regression outputs are unbounded.
- **(C) False.** Applying a sigmoid to linear regression outputs does not make it equivalent to logistic regression, since the optimization objective is still squared loss rather than cross-entropy loss.
- **(D) True.** Logistic regression corresponds to maximum likelihood estimation under a Bernoulli noise model, while linear regression assumes Gaussian noise.
- **(E) False.** Linear regression does not model labels as Bernoulli random variables; it models real-valued targets with Gaussian noise.

$$\boxed{\text{Correct options: (A), (B), (D)}}$$

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Solution to Q6

The dataset is

$$(x_1, y_1) = (2, 1), \qquad (x_2, y_2) = (-1, 0).$$

The model is

$$p(y = 1 \mid x) = \frac{1}{2} (1 + \tanh(wx)).$$

The average loss over the dataset is

$$L(w) = \frac{1}{2} \sum_{i=1}^{2} \ell_i(w), \quad \ell_i(w) = -[y_i \log p_i + (1 - y_i) \log(1 - p_i)].$$

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Step 1: Predictions at $w = 0$

Since $\tanh(0) = 0$,

$$p_i = \frac{1}{2} \quad \text{for both data points.}$$

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Step 2: Derivative of p w.r.t. w

$$\frac{dp}{dw} = \frac{1}{2}(1 - \tanh^2(wx))x.$$

At $w = 0$,

$$\frac{dp}{dw} = \frac{x}{2}.$$

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Step 3: Gradient of loss for one data point

For binary cross-entropy,

$$\frac{d\ell}{dw} = \left(\frac{1-y}{1-p} - \frac{y}{p}\right)\frac{dp}{dw}.$$

At $p = \frac{1}{2}$,

$$\frac{1-y}{1-p} - \frac{y}{p} = 2(1-y) - 2y = 2(1-2y).$$

Thus,

$$\frac{d\ell}{dw} = (1-2y)x.$$

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Step 4: Compute gradients

$$\frac{d\ell_1}{dw} = (1-2\cdot 1)\cdot 2 = -2,$$

$$\frac{d\ell_2}{dw} = (1-2\cdot 0)\cdot(-1) = -1.$$

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Step 5: Average gradient

$$\frac{dL}{dw} = \frac{1}{2}(-2-1) = -\frac{3}{2}.$$

$$\boxed{\nabla_w L(0) = -\frac{3}{2}}$$

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Solution to Q7

We want to maximize

$$f(x,y,z) = 2x + 4y + z^2$$

subject to the constraints

$$g_1(x,y,z) = x^2 + z^2 - 10 = 0, \qquad g_2(x,y,z) = y^2 + z^2 - 10 = 0.$$

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Step 1: Form the Lagrangian

Introduce Lagrange multipliers λ and μ:

$$\mathcal{L}(x,y,z,\lambda,\mu) = 2x + 4y + z^2 + \lambda(10 - x^2 - z^2) + \mu(10 - y^2 - z^2).$$

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Step 2: First-order conditions

Taking partial derivatives and setting them to zero:

$$\frac{\partial \mathcal{L}}{\partial x} = 2 - 2\lambda x = 0 \;\Rightarrow\; \lambda = \frac{1}{x},$$

$$\frac{\partial \mathcal{L}}{\partial y} = 4 - 2\mu y = 0 \;\Rightarrow\; \mu = \frac{2}{y},$$

$$\frac{\partial \mathcal{L}}{\partial z} = 2z - 2\lambda z - 2\mu z = 0 \;\Rightarrow\; z(1 - \lambda - \mu) = 0.$$

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Step 3: Solve cases

Case 1: $z = 0$. From the constraints,

$$x^2 = 10, \quad y^2 = 10.$$

To maximize f, take $x = \sqrt{10}$ and $y = \sqrt{10}$. Then

$$f = 2\sqrt{10} + 4\sqrt{10} = 6\sqrt{10} \approx 18.97.$$

Case 2: $z \neq 0$. Then

$$\lambda + \mu = 1.$$

Using $\lambda = \frac{1}{x}$ and $\mu = \frac{2}{y}$,

$$\frac{1}{x} + \frac{2}{y} = 1.$$

From the constraints,

$$x^2 = y^2 = 10 - z^2 \;\Rightarrow\; x = y = \sqrt{10 - z^2}.$$

Substitute into the multiplier equation:

$$\frac{1}{\sqrt{10 - z^2}} + \frac{2}{\sqrt{10 - z^2}} = 1 \;\Rightarrow\; \frac{3}{\sqrt{10 - z^2}} = 1.$$

Hence,

$$\sqrt{10 - z^2} = 3 \;\Rightarrow\; z^2 = 1.$$

Thus

$$x = 3, \quad y = 3, \quad z = \pm 1.$$

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Step 4: Evaluate the objective

$$f = 2(3) + 4(3) + 1 = 6 + 12 + 1 = 19.$$

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Conclusion

The maximum value of f under the given constraints is

$$\boxed{19},$$

achieved at

$$(x, y, z) = (3, 3, \pm 1).$$

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Solution to Q8

The odds ratio is

$$\frac{p(y = 1 \mid x)}{p(y = 0 \mid x)}.$$

- **(A) True.** The log-odds is

$$\log \frac{p}{1 - p} = w^\top x + b,$$

  which is linear in x.

- **(B) False.** Scaling (w, b) changes $w^\top x + b$ and hence changes the probabilities.

- **(C) True.** Adding a constant to b shifts the decision boundary.

- **(D) False.** Logistic regression models the log-odds as a linear function; probabilities are obtained via the sigmoid.

- **(E) True.** Increasing feature x_j by one unit increases the log-odds by w_j.

$$\boxed{\text{Correct options: (A), (C), (E)}}$$